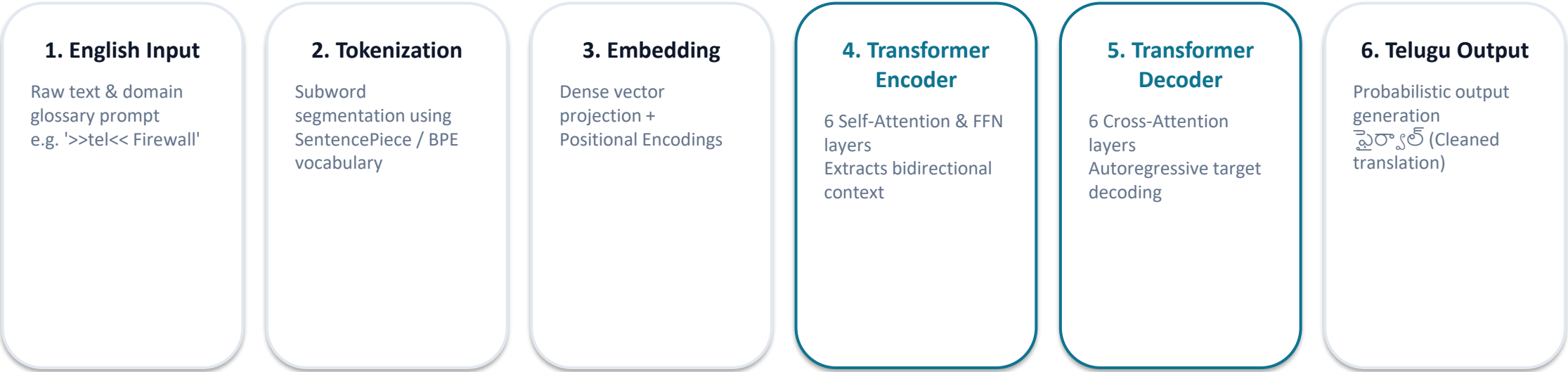


Architecture and Processing Pipeline of Small Language Models



Key Operational Principles of Small Language Models (SLMs):

- Focused Parameter Budget: SLMs (~50M–3B parameters) are trained for task-specific efficiency rather than open-ended multi-tasking.
- Compact Encoder-Decoder: Dual-stack Transformer architecture specializes in direct cross-lingual translation with minimal compute.
- High-Throughput Edge Inference: Consumes < 300 MB RAM, enabling instantaneous offline translation on standard consumer CPUs.

Recognized Compact Language Models and Production Deployments

Model Name	Year	Architecture & Characteristics	Organizations Using It	Limitations	Practical Products
MarianMT (Opus-MT)	2020	6L Enc - 6L Dec (~77M params) Seq2Seq NMT; lightweight CPU execution	Helsinki-NLP, Mozilla, Wikimedia	Restricted general reasoning; sensitive to domain shift	Firefox Offline Translations, Wikipedia tools
Microsoft Phi-3 Mini	2024	3.8B parameter decoder-only Trained on curated high- quality textbook data	Microsoft, Epic Systems, Azure AI	Cannot memorize vast factual knowledge bases	Copilot on-device, clinical note assistants
Google Gemma 2B	2024	2.6B parameter decoder Derived from Gemini research, lightweight KV	Google DeepMind, Android Developers	Limited multilingual vocabulary in base variant	Android AICore, on-device mobile AI
Apple OpenELM	2024	270M–3B parameter scaling Layer-wise non-uniform parameter allocation	Apple Machine Learning Research	Strict context window constraints	On-device Siri features, text prediction
DistilBERT / DistilBART	2019	6-layer distilled Transformer (~66M) Knowledge distillation from BERT/BART	Hugging Face, Grammarly, Bloomberg	Loss of ~3-5% nuance compared to teacher model	Real-time search indexing, edge NLP pipelines

Autoregressive Transformers & Resource Demands in Large Language Models

LLM Generation Pipeline:

1. Input Prompt: User query / instructions received.
2. Tokenization: Byte-Pair Encoding (tiktoken / Llama BPE).
3. Embedding & RoPE: High-dimensional projection + Rotary Position.
4. Deep Transformer Blocks: 32 to 128 stacked decoder layers.
5. Multi-Head Self-Attention: Dynamic Query-Key-Value (Q,K,V) relevance.
6. Feed Forward (MLP): Non-linear parameter transformation (SwiGLU).
7. Softmax & Next-Token Prediction: Generates token distribution.
8. Autoregressive Loop: Tokens appended sequentially to context.

Why LLMs Require High Computational Resources:

- **Massive Parameter Footprint:**
7B to 400B+ parameters require tens to hundreds of gigabytes of high-bandwidth VRAM just to store model weights in memory.
- **Quadratic Attention Complexity:**
Standard self-attention computes an $O(N^2)$ matrix across token sequences, causing exponential memory growth over long contexts.
- **KV-Cache Scaling:**
Storing Key and Value tensors during generation consumes vast memory for every concurrent user request.
- **Hardware Specialization:**
Requires clusters of dedicated AI accelerators (NVIDIA H100 / A100) with tensor parallelism and liquid cooling.

Comprehensive Dimensional Comparison Between Small and Large Models

Evaluation Dimension	Small Language Models (SLMs)	Large Language Models (LLMs)
Model Size (Disk)	~50 MB to 1.5 GB	15 GB to 800+ GB
Parameter Count	< 100M up to ~3B parameters	7B to 400B+ parameters
Hardware Requirements	Standard Laptop CPU / 1–2 GB RAM (No GPU needed)	High-end GPUs (A100/H100, 24GB–80GB+ VRAM)
Inference Speed	Fast (< 50–100 ms per token on standard CPU)	Requires high-compute GPU clusters & tensor parallelism
Cost & Energy	Near-zero runtime cost; energy-efficient on edge devices	Significant API pricing & high server electricity costs
Deployment Flexibility	Runs locally, on-device, offline, embedded systems	Predominantly cloud-hosted APIs / datacenter servers
Data Privacy & Security	100% private; data never leaves the local machine	Often sends sensitive text across external cloud networks
Typical Use Cases	Domain translation, glossary lookup, classification, IoT	Open-domain creative reasoning, complex coding, multi-turn chat

Cybersecurity Glossary Translation: English → Telugu Using MarianMT SLM

Pipeline Architecture:

Glossary PDF → PyMuPDF → English Terms → MarianMT SLM → Beam Search (k=4) → Artifact Cleaner → Telugu Glossary

Verified Model Output Samples (76.89M SLM):

- Firewall → ఫైర్వอลล์
- Authentication → ధృవీకరణ
- Authorization → ధృవీకరణ
- Digital Signature → డిజిటల్ సంతకం
- Password → సంకేతపదం
- Risk → ప్రమాదం
- Phishing → ఫిషింగ్

* Note: Beam search (k=4) and automated post-processing were used to reduce decoding artifacts. Specialized cybersecurity terminology remains a limitation of the general-domain model.

Cybersecurity Glossary: English -> Telugu Translation (MarianMT SLM)

Model: Helsinki-NLP/opus-mt-en-dra (~76.89M Parameters) | Marian Transformer | Inference: CPU

ID	English Cybersecurity Term	Telugu Translation (SLM Cleaned)
[01]	Access Control	యాక్సెస్ నియంత్రణ
[02]	Authentication	ధృవీకరణ
[03]	Authorization	ధృవీకరణ
[04]	Availability	ఎనలైయబిలిటీ
[05]	Confidentiality	ఎన్కోరింగ్
[06]	Integrity	యథార్థత
[07]	Non-repudiation	రేవరసేషన్
[08]	Cybersecurity	సైబర్ సెక్యూరిటీ
[09]	Cyber Attack	సైబర్ దండ
[10]	Data Breach	దత్తాంతం బ్రేక్
[11]	Encryption	ఎన్కోడింగ్
[12]	Decryption	ఎన్కోడింగ్
[13]	Cryptography	అర్థోగ్రాఫీ
[14]	Cryptographic Key	ముద్రాణ కీ
[15]	Digital Signature	డిజిటల్ సంతకం
[16]	Hash Function	హాష్ ఫంక్షన్
[17]	Firewall	ఫైర్వอลล์
[18]	Intrusion Detection System	ఇన్ట్రూజన్ డిటెక్షన్ సిస్టమ్

* Extracted from English_Cybersecurity_Glossary.pdf using PyMuPDF and translated with Beam Search (k=4)

Public Project Repository, Tech Stack & Reproducibility

Official GitHub Repository (Public):

<https://github.com/vignesh28102006/nlp-cybersecurity-slm-english-telugu>

Clone & Run: `git clone https://github.com/vignesh28102006/nlp-cybersecurity-slm-english-telugu.git`

Technologies & Libraries Used:

- Python 3.12: Core scripting & data processing runtime
- PyTorch: Tensor computation & CPU matrix acceleration
- Transformers: Hugging Face MarianMT model loading & tokenization
- PyMuPDF (fitz): High-performance PDF text & term extraction
- Pandas: Tabular dataset generation & CSV structuring
- Jupyter Notebook: Cell-by-cell demonstration & visualization

SLM Benchmark Summary:

- **Model Identifier:**
Helsinki-NLP/opus-mt-en-dra
- **Architecture:**
MarianMT Transformer (6L Enc / 6L Dec)
- **Exact Parameters:**
76,894,208 parameters (~76.89M)
- **RAM Footprint:**
293.3 MB (Well within 1 GB limit)
- **Load Time on CPU:**
1.27 seconds
- **Inference Speed:**
56 terms translated in 39.1 seconds (~698 ms/pair)